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Computing

“This ISA SPEC And Its COMPATIBILITY SUITES Will ALWAYS BE FREE”

RISC-V (pronounced ‘risk-five’) is an open instruction set architecture (ISA) based on reduced instruction set computing (RISC). What sets it apart from other ISAs is that, RISC-V ISA can be freely used for any purpose, permitting anyone to design, manufacture and sell RISC-V chips and software. While not the first open architecture ISA, it is significant because it is designed to be useful in a wide range of devices. Yunsup Lee, co-founder and chief technology officer at SiFive and Krste Asanovic, co-founder and chief architect at SiFive speak with Dilin Anand of *Electronics For You*

Q. What sparked the creation of RISC-V?

A. We wanted a simple, clean ISA that was efficient to implement and easy to extend to add specialised accelerators. We also wanted to be able to share with others. The tipping point was realising that we were wasting time adapting commercial ISAs to our then academic needs, and that we could never share the results once we were done.

The Indian angle. Indian Institute of Technology, Madras is developing six RISC-V open source CPU designs for six distinct uses, from small 32-bit CPUs for the Internet of Things (IoT) to large 64-bit CPUs for warehouse-scale computers such as server farms based on RapidIO and Hybrid Memory Cube technologies.

Q. What were some of the initial guidelines your founding team decided to set in stone, to ensure that RISC-V ecosystem scaled as it did?

A. First, we mandated that this ISA spec and compatibility suites will always be free.

Second, we froze the specification when we released version 2.0 in 2014 to provide stability.

Third, we created an independent non-profit foundation to provide a stable, long-lived home for the ISA.

Q. Among the various open source licences available, what was the line of reasoning used to select BSD for RISC-V?

A. We wanted to encourage and facilitate widespread use in the industry. We knew a non-permissive licence would have impeded the spread of the ISA and our implementations.

Q. What are the focus areas for RISC-V Foundation members over the next 12 months?

A. We are focused on improving RISC-V in the embedded space, adding faster interrupts, packed SIMD (single instruction, multiple data) extensions and providing a new embedded application binary interface, as well as adding vector instructions for artificial intelligence (AI) and standardising various parts of RISC-V security architecture.

Q. What were the initial bottlenecks faced as you set out to define the importance of an open ISA over pre-existing open source hardware?

A. Initially, many were under the impression that we were providing a single, free implementation rather than a broadly-usable spec. There are now many open source cores available, all compatible with a single spec.

Q. What is the plan to drive the community around RISC-V to grow and thrive across the world?

A. The Foundation now has more than 100 companies as members and is expanding its charter to help accommodate the demand, which is far greater than we ever anticipated three years ago when we started the foundation. It will launch many more events around the world, coordinating with governments to help with educational outreach, and developing research programmes and vocational training materials for practitioners.

The Foundation will also continue to extend ISA with new features and support for the rapid growth of the software ecosystem. **EFY**